Project: Worked examples
Title: Pipe pressure drop and pump duty
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

## 1. Scope and method

Friction and fitting losses through series pipe segments at a given flow, the static head, the system curve, the duty point against a pump curve entered as points, the affinity-law shift for a speed or impeller change, and the NPSH available at the pump suction.

| WHAT IT CALCULATES | RELATION |
|---|---|
| Mean velocity in the pipe | $v = Q/(\pi D^2/4)$ |
| Reynolds number, which decides laminar or turbulent flow | $Re = \rho v D/\mu$ |
| Friction factor for laminar flow (Re < 2300) | $f = 64/Re$ |
| Friction factor for turbulent flow, Colebrook–White by fixed-point iteration (at most 100 steps) from the Swamee–Jain value, which also cross-checks it | $1/\sqrt{f} = -2 \log10(\varepsilon/3.7D + 2.51/(Re\sqrt{f}))$ |
| Head lost in the pipe and its fittings | $h_f = f\,(L + L_{eq})/D \cdot v^2/2g + K\,v^2/2g$ |
| System curve, with friction scaled from the design flow as $Q^2$ | $H_{sys}(Q) = \Sigma dz + h_{out} + \Sigma h_f(Q)$ |
| Duty point, where the pump and system curves cross (found by bisection) | $H_{pump}(Q) = H_{sys}(Q)$ |
| Pump affinity laws for a change of speed n or impeller diameter D | $Q \propto n\,D^3$; $H \propto n^2\,D^2$ |
| Net positive suction head available, to compare with what the pump requires | $NPSH_a = p_{atm}/\rho g + z_s - h_{f,suction} - p_v/\rho g$ |

Basis: Flow of fluids through valves, fittings and pipe[1]; Internal flow: friction factor, minor losses, pump duty[2](Ch. 8); Flow of fluids in pipes and ducts (principles)[3](Section 4); Density of liquid water[4].

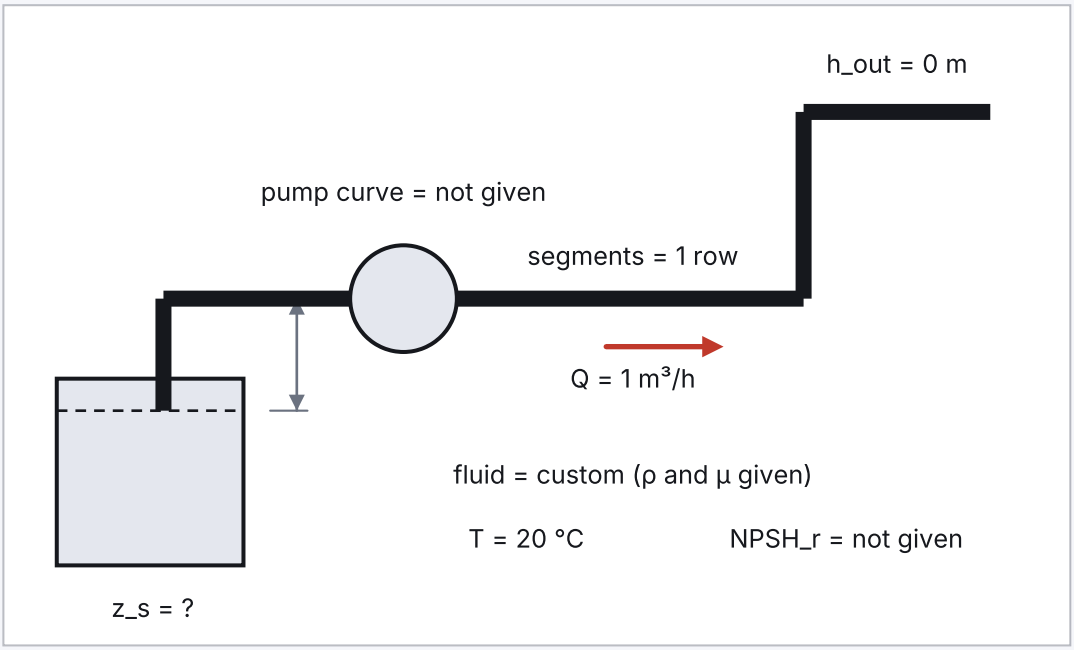



Figure 1 — A pump lifting from a suction level through the pipe segments as listed (three drawn) to an outlet that needs a pressure head. Not to scale; the labels are the inputs as entered.

## 2. Inputs

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| fluid | Fluid | custom (ρ and μ given) | | spec: Oil datasheet ISO VG 680 at 20 °C | any |
| T | Fluid temperature | 20 | °C | assumption: Ambient 20 °C. | 0 … 100 °C |
| rho_custom | Density (custom fluid) | 888 | kg/m³ | spec: Oil datasheet ISO VG 680 | 500 … 2000 kg/m³ |
| mu_custom | Dynamic viscosity (custom fluid) | 800 | mPa·s | spec: Oil datasheet ISO VG 680 at 20 °C | 0.1 … 1000 mPa·s |
| Q | Design flow rate | 1 | m³/h | spec: Transfer pump duty | 0.01 … 100000 m³/h |
| segments | Pipe segments in series | 1 row(s) | | drawing: DRG-L-020 rev A | any |
| h_out | Pressure head required at the outlet | 0 | m | assumption: Open discharge. | 0 … 1000 m |
| v_max | Velocity limit (warning) | 3 | m/s | assumption: Practice limit. | 0.5 … 10 m/s |
| n_ratio | Speed ratio $n_2/n_1$ (affinity) | 1 | | default | 0.2 … 2 |
| D_ratio | Impeller diameter ratio $D_2/D_1$ (affinity) | 1 | | default | 0.7 … 1.1 |
| eta_pump | Pump efficiency at duty (for shaft power) | 0.7 | | default | 0.2 … 0.95 |
| p_atm | Atmospheric pressure (NPSH) | 101.325 | kPa | default | 50 … 110 kPa |

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| z_s | Static head at the pump suction (+ flooded, − lift) | 0 | m | default | -10 … 100 m |

| TABLE ROW | VALUES |
|---|---|
| segments[1] | name = transfer line; L = 40 m; D = 50 mm; eps = 0 mm; K = 0; L_eq = 0 m; dz = 0 m; side = discharge |

## 3. Assumptions

A1. Steady, incompressible, fully developed flow; all segments carry the full design flow in series. (Darcy–Weisbach.)
A2. Fitting losses scale with velocity head (K method) or add pipe length (equivalent-length method); both may be used on one segment. (Crane TP-410 practice.)
A3. The system curve scales friction losses with $Q^2$ from the design flow (turbulent assumption); the static part is independent of flow. ($H_{sys} = H_{static} + k\,Q^2$.)
A4. Water properties are from published correlations at atmospheric pressure (0–100 °C); any other fluid is entered with its own density and viscosity and their provenance. (Kell 1975; Vogel fit; Hyland & Wexler 1983.)
A5. T (Fluid temperature) is taken as 20 degC. (Ambient 20 °C.)
A6. h_out (Pressure head required at the outlet) is taken as 0 m. (Open discharge.)
A7. v_max (Velocity limit (warning)) is taken as 3 m/s. (Practice limit.)

## 4. Working

[5] rho = rho_custom
= 888 kg/m³
= 888 kg/m³
Density (as entered)
[6] mu = mu_custom
= 800 mPa·s
= 800 mPa·s
Dynamic viscosity (as entered)
[10] v_s1 = transfer line: Q / (π D²/4)
= 1 m³/h / (π × 50² mm² / 4)
= 0.14147 m/s
Velocity, transfer line
[11] Re_s1 = transfer line: ρ v D / μ
= 888 × 0.14147 × 0.05 / 0.8
= 7.8516
Reynolds number, transfer line
[12] f_s1 = transfer line: Darcy friction factor
= 64 / Re (laminar)
= 8.1512
Friction factor, transfer line
[13] h_f_s1 = transfer line: f (L + L_eq)/D · v²/2g + K v²/2g
= 8.1512 × (40 + 0) / 0.05 × 0.0010204 + 0 × 0.0010204
= 6.6542 m
Head loss, transfer line
[14] h_f_total = Σ h_f
= 1 segment(s)
= 6.6542 m
Total friction and fitting loss at the design flow

```
[15] H_static = Σ dz + h_out
= 0 + 0
= 0 m
```
Static head
```
[16] H_sys = H_static + h_f_total
= 0 m + 6.6542 m
= 6.6542 m
```
System head at the design flow
```
[17] dp_total = rho * 9.80665 m/s^2 * h_f_total
= 888 kg/m³ × 9.80665 m/s^2 × 6.6542 m
= 57.947 kPa
```
Friction pressure drop at the design flow
```
[18] dp_sys = rho * 9.80665 m/s^2 * H_sys
= 888 kg/m³ × 9.80665 m/s^2 × 6.6542 m
= 57.947 kPa
```
System pressure at the design flow
```
[19] P_hyd = rho * 9.80665 m/s^2 * Q * H_sys
= 888 kg/m³ × 9.80665 m/s^2 × 1 m³/h × 6.6542 m
= 0.016096 kW
```
Hydraulic power at the design point
```
[22] v_peak = highest segment velocity
= 0.14147 m/s
= 0.14147 m/s
```
Highest velocity
```
[23] u_v = v_peak / v_max
= 0.14147 / 3
= 0.047157
```
Velocity utilisation

## 5. Results

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| v_s1 | 0.14147 m/s | Velocity, transfer line |
| Re_s1 | 7.8516 | Re, transfer line |
| f_s1 | 8.1512 | f, transfer line |
| h_f_s1 | 6.6542 m | h_f, transfer line |
| h_f_total | 6.6542 m | Σ h_f |
| H_static | 0 m | Static head |
| H_sys | 6.6542 m | System head at Q |
| dp_total | 57.947 kPa | Friction Δp |
| dp_sys | 57.947 kPa | System Δp |
| P_hyd | 0.016096 kW | Hydraulic power |
| v_peak | 0.14147 m/s | Highest velocity |

## 6. Checks (governing first)

| RESULT | CHECK | DEMAND | CAPACITY | UTILISATION | REF |
|---|---|---|---|---|---|
| PASS | Velocity ≤ limit | u_v = 0.047157 | u_v ≤ 1 | 0.0472 | |

## 7. Applicability

WARNING — input.n_ratio: n_ratio (Speed ratio $n_2/n_1$ (affinity)) was not entered; the template default 1 is used.
WARNING — input.D_ratio: D_ratio (Impeller diameter ratio $D_2/D_1$ (affinity)) was not entered; the template default 1 is used.
WARNING — input.eta_pump: eta_pump (Pump efficiency at duty (for shaft power)) was not entered; the template default 0.7 is used.
WARNING — input.p_atm: p_atm (Atmospheric pressure (NPSH)) was not entered; the template default 101.3 kPa is used.
WARNING — input.z_s: z_s (Static head at the pump suction (+ flooded, − lift)) was not entered; the template default 0 m is used.
WARNING — laminar: segments: transfer line is laminar (Re = 7.8516); f = 64/Re.

## 8. Verification

```
Template bs.pipe-pump@1.0.1 Content hash 332014d4a12dcd6a Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
```

## 9. References

1. Crane Co. Flow of Fluids Through Valves, Fittings and Pipe. Technical Paper No. 410. Crane Co.
2. Çengel YA, Cimbala JM. Fluid Mechanics: Fundamentals and Applications. 4th ed. McGraw-Hill Education; 2018.
3. Chartered Institution of Building Services Engineers. CIBSE Guide C: Reference Data. CIBSE; 2007.
4. Kell GS. Density, thermal expansivity, and compressibility of liquid water from 0° to 150°C: correlations and tables for atmospheric pressure and saturation reviewed and expressed on 1968 temperature scale. J Chem Eng Data. 1975;20(1):97-105.